School of Engineering MSc Robotics AERO62520 Robotic Systems Design Project

Final Design Review

Objective
The purpose of this activity is to conduct a final review of your design to make sure it meets the requirements specified in the 'Design Requirements Analysis' coursework. In the following descriptions, the word 'robot' means the mobile manipulator (Leo Rover + Manipulator) and all its peripherals.

The core of the Final Design Review coursework should have the same structure as the Preliminary Design Review (using the original document as the starting point). You should provide an updated version with an introduction describing the modifications and changes you have made. You are also requested to write new sections in form of sustainability checklist and cyber security considerations (see details below). All sections should have appropriate contextualisation text (i.e., don't just put the images in).

- **Introduction (50%)**
  - A summary of the problem statement and objectives.
  - A summary of how you have addressed feedback from the PDR **(25%)**
  - A summary of any modifications you have made since the PDR **(25%)**
- **Sustainability Checklist (20%)**
  - Use the template provided to indicate how sustainability has been considered within the design. The template is provided on blackboard.
- **Cyber Security Considerations (5%)**
  - A short section describing any cyber security considerations.
- **Updated Preliminary Design Review (25%)**
  - **System**
    - System block-diagram of all the components in your robot.
      - Ensure all of the components that have been used are described and justified as to why they are needed.
  - **Mechanical Design**
    - 3D CAD model of your robot.
    - Design files for your payload sled.
      - Discussion of any design trade-offs or challenges with component placement.
      - Discussion about manufacturability and suitability of your sled design (e.g. strength, stability, etc…)
  - **Electrical Design**
    - Power connection diagram for all the peripherals on your robot.
      - This could be included as part of the system block diagram, but it should include values and units
    - Power Budget
      - Calculate if you have enough power to run everything.
  - **Software Design**
    - RQT Graph with statistics.
      - There should be a clear link between the RQT Graph and the System block diagram.
    - A link to your Team's Git repository
      - Documentation of your code repository will be evaluated.
  - **Project Plan**
    - A project plan (Gantt Chart) with clear work packages, resource allocations (who is doing what), milestones and deliverables for S2.

    - **Analysis**
        - Using the Requirements Verification Matrix, you should analyse your design meets the requirements.
- **Peer Review**
    - All students must complete the Peer Review assessment on Blackboard.

Submission
There will be an assignment submission link on Canvas. This assessment is a 'Group' assessment, however the mark will be scaled by the Peer Review 3 weighting. **One student from each Team must complete the assignment by the deadline.**

The submission deadline is **Friday 20th March 2026 14:00**.

Mitigating Circumstances
Any mitigating circumstance requests should be directed to the Teaching Office.

SAW & PL
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